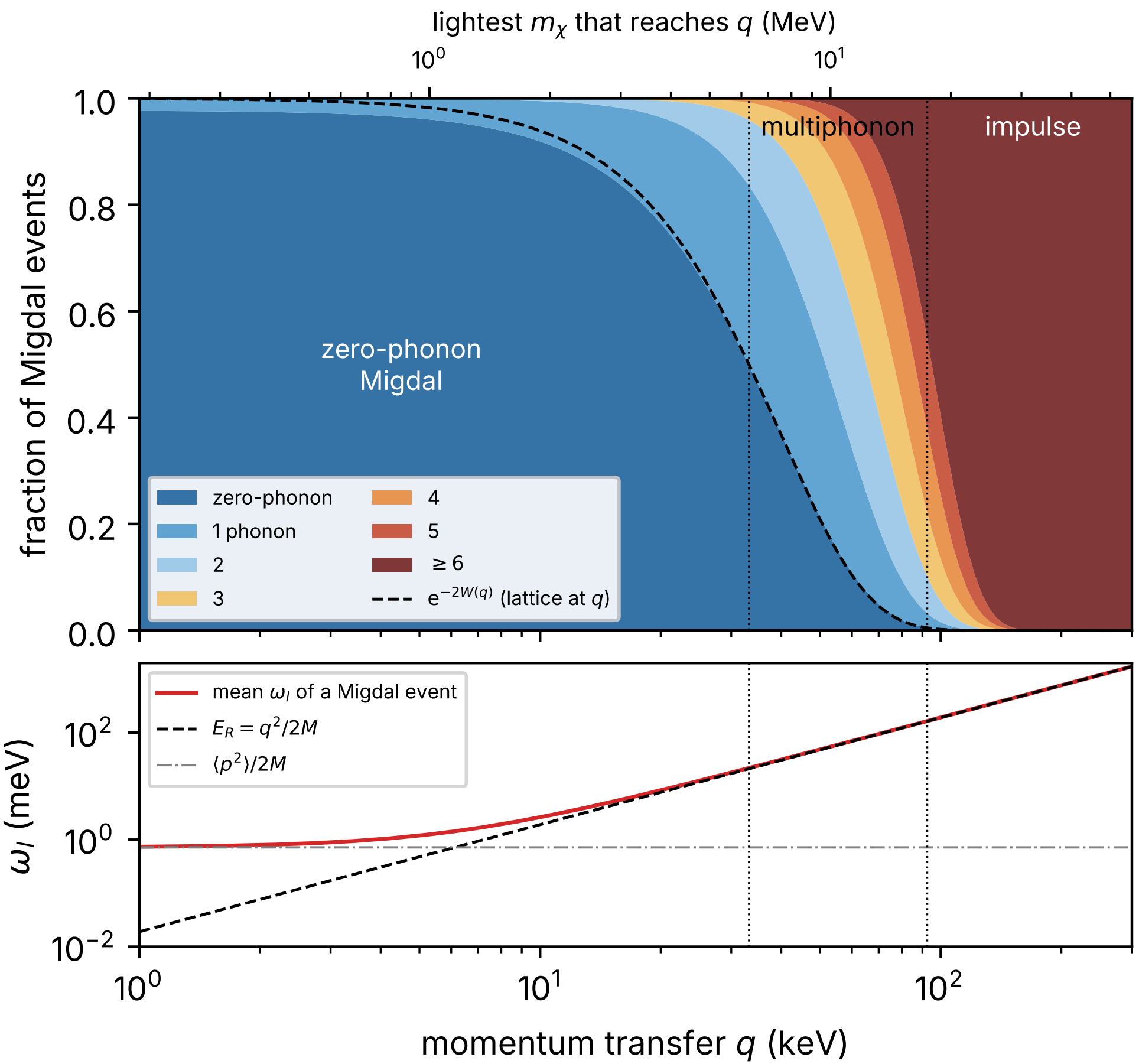
lightest m_χ that reaches q (MeV)
10^0
10^1
1.0
0.8
0.6
0.4
0.2
0.0
fraction of Migdal events
multiphonon
impulse
zero-phonon
Migdal
zero-phonon
1 phonon
2
3
4
5
≥ 6
$e^{-2W(q)}$ (lattice at q)
mean ω_I of a Migdal event
$E_R = q^2/2M$
$\langle p^2 \rangle/2M$
10^2
10^0
10^{-2}
ω_I (meV)
10^0
10^1
10^2
momentum transfer q (keV)